

# Qatar-2b

R-2203

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-2\_120206 · created 2026-08-14T00:54:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2455963.89411 +/- 0.00048 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.184 +/- 0.04                    |
| Transit depth (Rp/Rs)^2                           | 3.4 +/- 1.49 [%]                  |
| Orbital Inclination (inc)                         | 89.66 +/- 1.64                    |
| Airmass coefficient 1 (a1)                        | 0.932 +/- 0.028                   |
| Airmass coefficient 2 (a2)                        | 0.023 +/- 0.013                   |
| Scatter in the residuals of the lightcurve fit is | 3.02 %                            |
| Best Comparison Star                              | #2 - [32, 68]                     |
| Optimal Method                                    | PSF photometry                    |
| Transit Duration (day)                            | 0.0777 +/- 0.0032                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R* fit vs published    | 0.184 ± 0.04 vs 0.16327 (deviation 0.52σ) |
| Duration fit vs published | 0.0777 d vs 0.0767 d (deviation 0.32σ)    |
| Mid-time O–C              | -0.9 min                                  |
| Depth SNR                 | 4.6                                       |
| Residual scatter          | 3.02 %                                    |

## Light curve

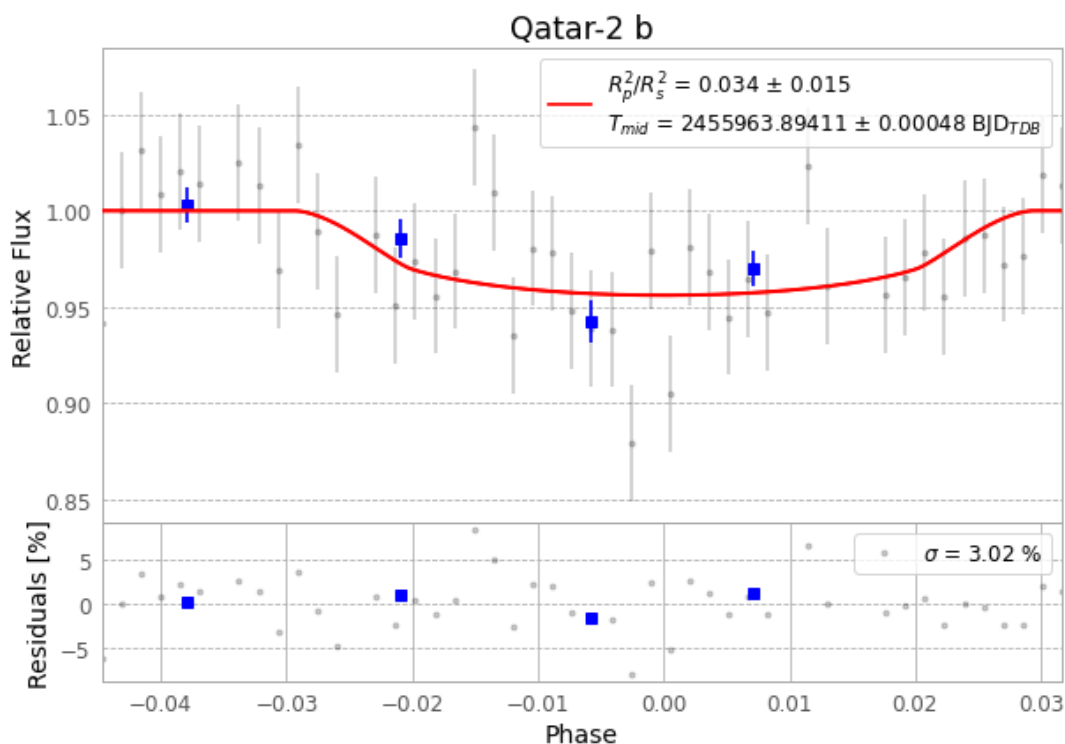


Figure 1 — FinalLightCurve\_Qatar-2 b\_2012-02-06.png (SHA-256 1bc20135fdbac969...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [217, 212], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 24f16280598d1057...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1dc3d22

## Data provenance

47 FITS frames (31 MB), Qatar - 2120206075713.FITS → Qatar - 2120206102413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-120206.log (8c4b8001769e...), FinalLightCurve\_Qatar-2 b\_2012-02-06.png (1bc20135fdbac...), FinalLightCurve\_Qatar-2 b\_2012-02-06.csv (7281f55fff23...)

## Compute resources

Wall time 97.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:54Z.